

Marketing Mix Model (MMM) Methodology Whitepaper

Auto-Modeller Bayesian MMM Platform

1. Purpose of the Model

The Marketing Mix Model (MMM) estimates the **incremental impact of marketing activities on business outcomes** using aggregated time-series data.

The system is designed to:

- measure marketing effectiveness
 - estimate channel ROI
 - quantify marketing contribution
 - understand diminishing returns
 - support strategic marketing decisions
-

2. Data Requirements

The model requires **historical time series data**.

Data Format

date | region | sub_brand | variable | value

Example:

date	region	sub_brand	variable	value
2023-01-01	US	BrandA	clm_call	150
2023-01-01	US	BrandA	volume	5200

3. Variable Categories

Variables are categorized in **info.csv**.

KPI

Primary outcome variable.

Examples:

volume
sales
prescriptions

Media Variables

Marketing activities measured as **exposure variables**.

Examples:

clm_call
non_clm_call
mass_email
email_1to1
phone_call
webinar

Spend is used **only for ROI calculation**, not modelling.

Control Variables

External drivers.

Examples:

competitor_activity
economic_indicator
holiday_index

4. Model Structure

The model predicts KPI:

$$Y_{t,g,b} = Base_{t,g,b} + Media_{t,g,b} + Control_{t,g,b} + \epsilon_{t,g,b}$$

Where

term	description
------	-------------

Base	baseline demand
------	-----------------

Media	marketing impact
-------	------------------

term	description
------	-------------

Control external drivers	
--------------------------	--

ε	residual noise
---------------	----------------

5. Baseline Component

Baseline demand includes:

intercept

trend

seasonality

Intercept

$$\alpha \sim Normal(0,1)$$

Represents baseline demand level.

Trend

$$Trend_t = \beta_{trend} \cdot t$$

Where

$$\beta_{trend} \sim Normal(0,0.1)$$

Captures market growth or decline.

Seasonality

Weekly seasonal effects are modeled using Fourier terms.

$$Season_t = \alpha_1 \sin(2\pi t/52) + \alpha_2 \cos(2\pi t/52)$$

6. Media Effect Modeling

Marketing effects undergo two transformations:

- 1 **Adstock** (carryover)
 - 2 **Saturation** (diminishing returns)
-

7. Adstock Transformation

Marketing impact persists beyond the week it occurs.

Adstock equation:

$$Adstock_t = X_t + \theta Adstock_{t-1}$$

Where

parameter meaning

θ retention rate

Prior

$$\theta \sim Beta(2,2)$$

Bounded between:

$$0.01 \leq \theta \leq \theta_{\text{cap}}$$

8. Saturation Transformation

Marketing impact exhibits diminishing returns.

Hill-type function:

$$S(X) = \frac{X}{\gamma + X}$$

Where

parameter meaning

γ saturation level

Small $\gamma \rightarrow$ faster saturation.

Prior

$$\gamma \sim HalfNormal(\sigma)$$

9. Media Contribution

Media contribution is computed as:

$$Media_t = \sum_i \beta_i \cdot S(Adstock(X_{i,t}))$$

Where

parameter meaning

β_i media effectiveness

10. Funnel Hierarchy Constraint

Marketing channels are categorized as:

lower funnel

mid funnel

upper funnel

Coefficients follow ordering:

$$\beta_{lower} \leq \beta_{mid} \leq \beta_{upper}$$

This stabilizes estimation and aligns with marketing theory.

11. Control Variables

Control effects enter linearly.

$$Control_t = \sum_j \beta_j Z_{j,t}$$

Prior:

$$\beta_j \sim Normal(0,0.5)$$

12. Observation Model

Final KPI distribution:

$$Y_t \sim Normal(\mu_t, \sigma)$$

Where

$$\mu_t = Base_t + Media_t + Control_t$$

13. Bayesian Estimation

The model is estimated using **Hamiltonian Monte Carlo**.

Sampling configuration:

chains = 4

draws = 1000

tune = 1000

target_accept = 0.95

14. Contribution Extraction

Media contributions are extracted using a **delta method**.

Contribution:

$$Contribution_{i,t} = \beta_i \cdot S(Adstock(X_{i,t}))$$

Total channel contribution:

$$Contribution_i = \sum_t Contribution_{i,t}$$

15. Revenue Calculation

If KPI = volume:

$$Contribution_{sales} = Contribution_{volume} \times revenue_per_unit$$

16. ROI Calculation

$$ROI_i = \frac{Contribution_{sales,i}}{Spend_i}$$

17. Effectiveness Metric

Channel efficiency:

$$Effectiveness_i = \frac{ContributionShare_i}{SpendShare_i}$$

Interpretation:

value meaning

>1 under-invested

≈1 balanced

<1 overspending

18. Response Curves

Response curves represent KPI as a function of spend.

$$Revenue = f(Spend)$$

These curves are used for:

- marginal ROI
 - diminishing returns analysis
 - investment planning
-

19. ROI Stability Diagnostic

Uncertainty ratio:

$$Stability = \frac{ROI_{p95} - ROI_{p05}}{|ROI|}$$

Classification:

value classification

- <1 stable
- 1-2 moderate
- >2 unstable

20. Model Diagnostics

Model quality metrics:

metric	description
RMSE	prediction error
MAPE	relative error
R ²	variance explained
Bayesian R ²	posterior model fit

21. Model Assumptions

Key assumptions:

1. Media effects are additive.
2. Carryover captured by adstock.
3. Diminishing returns captured by saturation.
4. Residual errors follow a normal distribution.
5. External variables sufficiently capture non-marketing drivers.